4.6 Overlapping Triangles & Special Segments

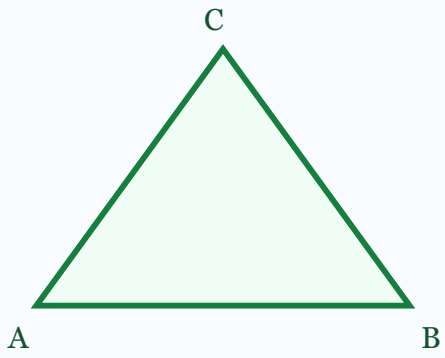
Triangles have four key special segments. A **median** connects a vertex to the midpoint of the opposite side. An **altitude** is a perpendicular segment from a vertex to the line containing the opposite side (in an obtuse triangle, the altitude from an acute vertex can land outside the triangle). An **angle bisector** splits a vertex angle into two congruent angles. A **perpendicular bisector** is perpendicular to a side at its midpoint, and the **Perpendicular Bisector Theorem** guarantees every point on it is equidistant from the side's two endpoints. The three medians of a triangle always meet at the **centroid**, which divides each median in a $2:1$ ratio from the vertex, and can be found by averaging the three vertices' coordinates. When triangles overlap (sharing a side, an angle, or lying across intersecting lines), a good first step is to redraw them separately, then look for shared parts via the Reflexive Property or the Vertical Angles Theorem. This worksheet combines overlapping-triangle proofs with numeric and coordinate work on medians, altitudes, bisectors, and centroids.

Examples

Example 1: Overlapping triangles with parallel sides

$\overline{AC}$ and $\overline{BD}$ intersect at E, with $\overline{AB} \parallel \overline{CD}$ and $AE = CE$. Prove $\triangle ABE \cong \triangle CDE$.

Solution



Redrawing the two overlapping triangles separately makes the shared parts easier to see.

Since $\overline{AB} \parallel \overline{CD}$ and $\overline{AC}$ is a transversal, alternate interior angles are congruent: $\angle BAE \cong \angle DCE$.

$\angle AEB$ and $\angle CED$ are vertical angles (formed by intersecting segments $\overline{AC}$, $\overline{BD}$), so $\angle AEB \cong \angle CED$ by the Vertical Angles Theorem.

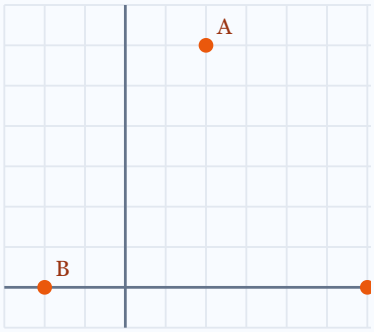
$\overline{AE} \cong \overline{CE}$ is given, and it is included between $\angle BAE$ and $\angle AEB$ (likewise $\overline{CE}$ between $\angle DCE$ and $\angle CED$).

Conclusion: $\triangle ABE \cong \triangle CDE$ by ASA.

Example 2: Median length by coordinates

Triangle has vertices $A(2, 6)$, $B(-2, 0)$, $C(6, 0)$. Find the length of the median from A.

Solution



The median from A goes to the midpoint of $\overline{BC}$.

Find the midpoint M of $\overline{BC}$ (the foot of the median): $M = (\frac{-2+6}{2}, \frac{0+0}{2}) = (2, 0)$.

Apply the distance formula from $A(2, 6)$ to $M(2, 0)$:
$AM = \sqrt{(2-2)^2 + (6-0)^2} = \sqrt{0+36} = 6$.
Answer: median $\overline{AM} = 6$.

Example 3: Altitude outside an obtuse triangle

In obtuse $\triangle ABC$ with the obtuse angle at B, where does the altitude from A meet line BC?

Solution

The altitude from A must be perpendicular to line BC.
Because $\angle B$ is obtuse (greater than $90°$), the perpendicular dropped from A cannot land between B and C — it lands beyond B, on the extension of $\overline{CB}$.
Conclusion: the foot of the altitude lies outside segment $\overline{BC}$; two of the three altitudes of an obtuse triangle fall outside the triangle.

Example 4: Angle bisector algebra

$\overline{BD}$ bisects $\angle ABC$, with $\angle ABD = 3x + 4$ and $\angle DBC = 5x - 10$. Find x, each half-angle, and the full angle $\angle ABC$.

Solution

Since $\overline{BD}$ bisects $\angle ABC$, the two halves are congruent:

$3x + 4 = 5x - 10$

$14 = 2x \Rightarrow x = 7$.

Each half: $\angle ABD = 3(7) + 4 = 25°$ (check: $5(7) - 10 = 25$ ✓).

Full angle: $\angle ABC = 25 + 25 = 50°$.

Answer: $x = 7$; each half $= 25°$; $\angle ABC = 50°$.

Example 5: Perpendicular Bisector Theorem

Point P lies on the perpendicular bisector of $\overline{AB}$, which meets $\overline{AB}$ at midpoint M. Prove $PA = PB$.

Solution

Identify the given information: $\overline{PM}$ is perpendicular to $\overline{AB}$, so $\angle PMA = \angle PMB = 90°$; and M is the midpoint, so $AM = BM$.

$\overline{PM} \cong \overline{PM}$ by the Reflexive Property (shared leg).

$\triangle PMA$ and $\triangle PMB$ are right triangles with two congruent legs ($PM = PM$, $AM = BM$) and the right angle included between them.

By SAS, $\triangle PMA \cong \triangle PMB$.

Conclusion: $PA = PB$ by CPCTC — every point on the perpendicular bisector is equidistant from A and B.

Example 6: Centroid ratio

Median $\overline{AM}$ of a triangle has length 18. The centroid G lies on $\overline{AM}$. Find AG and GM.

Solution

The centroid divides every median in a $2:1$ ratio, measured from the vertex.

So $AG:GM = 2:1$, meaning $AG = \frac{2}{3}(18)$ and $GM = \frac{1}{3}(18)$.

$AG = 12$ and $GM = 6$.

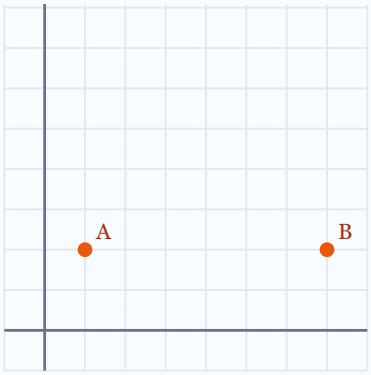
Check: $AG + GM = 12 + 6 = 18 = AM$ ✓.

Answer: $AG = 12$, $GM = 6$.

Example 7: Centroid from coordinates

Triangle has vertices $A(1, 2)$, $B(7, 2)$, $C(4, 11)$. Find the coordinates of the centroid.

Solution



The centroid is the average of the three vertices.

The centroid is found by averaging the x-coordinates and averaging the y-coordinates of all three vertices:

$$G_x = \frac{1 + 7 + 4}{3} = \frac{12}{3} = 4$$

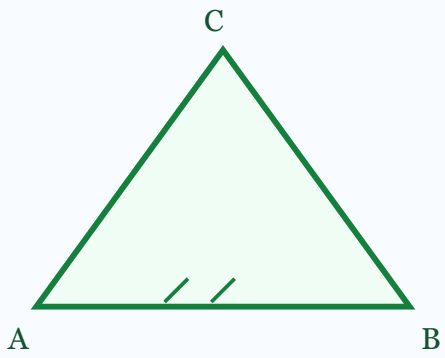
$$G_y = \frac{2 + 2 + 11}{3} = \frac{15}{3} = 5.$$

Answer: $G(4, 5)$.

Example 8: When median equals altitude

In $\triangle ABC$, D is the midpoint of $\overline{BC}$ (so $\overline{AD}$ is a median), and also $\overline{AD} \perp \overline{BC}$ (so $\overline{AD}$ is an altitude). Prove $AB = AC$.

Solution



$\overline{AD}$ is both a median and an altitude — it will force the triangle to be isosceles.

Since $\overline{AD}$ is an altitude, $\angle ADB = \angle ADC = 90°$ (right angles).
Since $\overline{AD}$ is a median, D is the midpoint of $\overline{BC}$, so $BD = DC$.
$\overline{AD} \cong \overline{AD}$ by the Reflexive Property (shared leg).
$\triangle ABD$ and $\triangle ACD$ are right triangles with two congruent legs and the right angle included — congruent by SAS (or HL, using AD as the shared leg).
Conclusion: $AB = AC$ by CPCTC — the triangle must be isosceles.

Example 9: Overlapping triangles — redraw first

$\triangle ABC$ and $\triangle ADC$ overlap, sharing side $\overline{AC}$, with $AB = AD$ and $\angle BAC \cong \angle DAC$. Prove $\triangle ABC \cong \triangle ADC$, and describe a helpful first step.

Solution

A helpful first step is to redraw $\triangle ABC$ and $\triangle ADC$ separately, so the shared side $\overline{AC}$ is easier to track in each copy.
List the given parts: $AB = AD$ (given), $\angle BAC \cong \angle DAC$ (given), $\overline{AC} \cong \overline{AC}$ (Reflexive Property, shared side).
Two sides and the included angle are congruent.
Conclusion: $\triangle ABC \cong \triangle ADC$ by SAS.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Shared side

Two overlapping triangles share a side. Which property justifies the side is congruent to itself?

Answer: **Verification:**

Problem 2: Shared angle

Two overlapping triangles share an angle. Which property justifies the angle is congruent to itself?

Answer: **Verification:**

Problem 3: Median

A median connects a vertex to what point of the opposite side?

Answer: **Verification:**

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Problem 4: From a median

$\overline{AD}$ is a median to $\overline{BC}$. What must be true about BD and DC?

Answer: **Verification:**

Problem 5: Altitude

An altitude from a vertex is perpendicular to what?

Answer: **Verification:**

Problem 6: Right angles

An altitude always creates angles measuring what?

Answer: **Verification:**

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Problem 7: Angle bisector

An angle bisector splits an angle into two angles that are what?

Answer: **Verification:**

Problem 8: Overlap strategy

What is a helpful first step when triangles overlap in a diagram?

Answer: **Verification:**

Problem 9: HL overlap

Two right triangles share a leg and have congruent hypotenuses. Which postulate applies?

Answer: **Verification:**

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Problem 10: Obtuse altitude

In an obtuse triangle, where can an altitude from an acute vertex fall?

Answer: **Verification:**

Problem 11: Centroid ratio

The centroid divides each median in what ratio, measured from the vertex?

Answer: **Verification:**

Problem 12: Perpendicular bisector

A point on the perpendicular bisector of a segment is always equidistant from what?

Answer: **Verification:**

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Problem 13: Challenge — angle bisector algebra

$\overline{BD}$ bisects $\angle ABC$, with $\angle ABD = (4x - 3)°$ and $\angle DBC = (2x + 11)°$. Find x, each half-angle, and $\angle ABC$.

Answer: **Verification:**

Problem 14: Challenge — median and isosceles test

Triangle has vertices $A(0, 8)$, $B(-6, 0)$, $C(6, 0)$. Find the median $\overline{AM}$ from A (to the midpoint of $\overline{BC}$), determine if it is also an altitude, and state what that implies about the triangle.

Answer: **Verification:**

Problem 15: Challenge — centroid verification

Triangle has vertices $A(3, 9)$, $B(-3, 0)$, $C(9, 0)$. Find the centroid G, the midpoint M of $\overline{BC}$, the median length AM, and verify G divides $\overline{AM}$ in ratio $2:1$ from A.

Answer: **Verification:**

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✓ Answer Key

1. Reflexive Property
2. Reflexive Property
3. The midpoint
4. $BD = DC$
5. The opposite side (or the line containing it)
6. $90°$
7. Congruent
8. Redraw the triangles separately
9. HL
10. Outside the triangle, on the extension of the opposite side
11. $2 : 1$
12. The segment's two endpoints
13. $x = 7$; each half $= 25°$; $\angle ABC = 50°$
14. $AM = 8$; since $AM \perp BC$, it is also an altitude, so $\triangle ABC$ is isosceles ($AB = AC = 10$)
15. $G(3, 3)$; $M(3, 0)$; $AM = 9$; $AG = 6$, $GM = 3$ (ratio $2 : 1$) ✓

Great work — keep practising!